Ch 07: Locking (Concurrency I)

xv6 a simple Unix-like teaching Operating System

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1 Races and Locks

2 Title



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Concurrency (1 of 2)

- One source of interleaving is multiprocessor hardware
  - Modern OS, and also xv6, allow kernel execution on all CPUs in multicore hardware
  - Multiple CPU architecture share physical RAM, OS exploits sharing in maintaining kernel data structures that all CPUs read and write
- Even without multiple processors, kernel interleaves execution of code (use and kernel)
  - System calls interleave user process to access data structures
  - Interrupts, device handlers can modify same kernel structures as interruptable code

Concurrency

Concurrency refers to the situations in which multiple instruction streams are interleaved, due to multiprocessor parallelism, thread switching, or interrupts.



Concurrency (2 of 2)

Concurrency or interleaving raises the possibility of one thread/CPU reading a data structure while another thread/CPU is mid-way through updating it, or even multiple thread/CPUs updating the same data simultaneously.

Without careful design, such parallel (concurrent) access is likely to yield incorrect results or a broken data structure.



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Kernel Design and Concurrency

- Kernels are full of concurrently-accessed data
  - For example, in xv6 two CPUs (or interleaved threads) could simultaneously call `kalloc()` thereby concurrently popping from the head of the free list.
  - **Pros:** like to allow for lots of concurrency as this yields increased performance through parallelism, thus increasing OS responsiveness
  - **Cons:** kernel designers must convince themselves of correctness of kernel code despite such concurrency (hard)

Strategies aimed at correctness under concurrency, and abstractions that support them, are called **concurrency control** techniques.



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Locks

Lock

Lock: A lock provides mutual exclusion, ensuring that only one CPU/thread at a time can hold the lock.

- If associate a lock with each shared data item and code always holds associated lock when accessing the item, ensures only one CPU/thread at a time
- Although locks are an easy-to-understand concurrency control mechanism, the downside of locks is that they can limit performance, because they **serialize** concurrent operations.

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Summary





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